Biposha
23BCS10735
EP-801-A
TOC

Average learner

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Ques Given the DFA accepting the lang. over the alphabet {0,1} that have the set of all strings that either begins or end (or both) with 01.

Ans $\Sigma = \{0,1\}$

The lang. consists of all string with 01 or end with 01.

state $q_0 \rightarrow$ Initial
$q_1 \rightarrow$
q_2
q_3
q_4

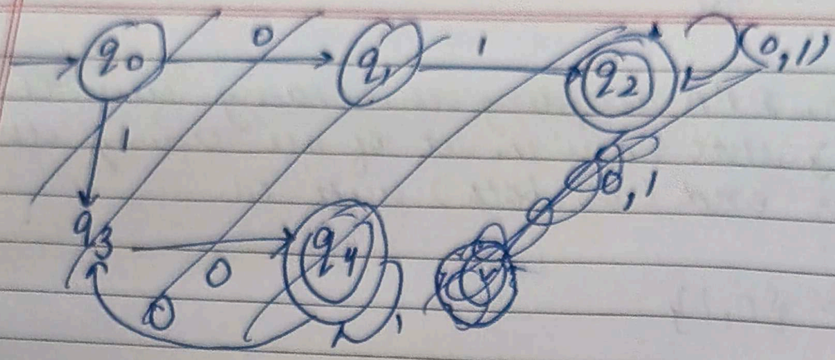
final states $F = \{q_2, q_4\}$

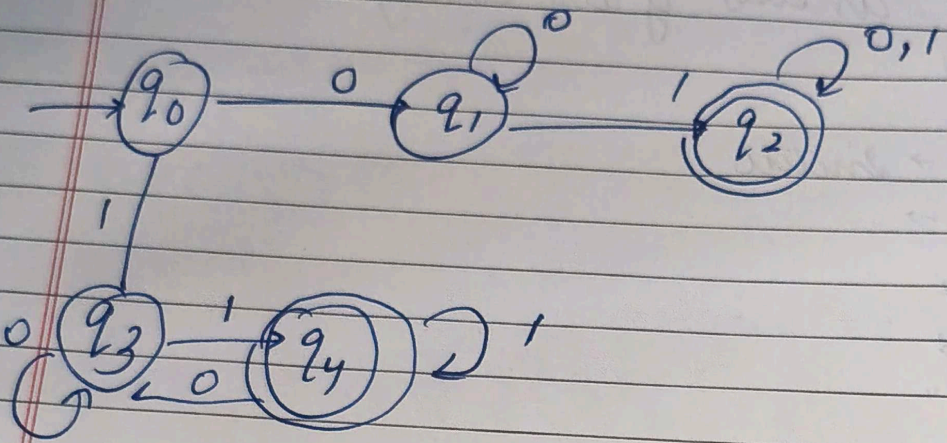
Transition Table.

q present	Input 0	Input 1
q_0	q_1	q_3
q_1	q_1	q_2
q_2	q_2	q_2
q_3	q_3	q_4
q_4	q_4	q_4

Thus, the DFA accepts string that begin with 01 or end with 01.

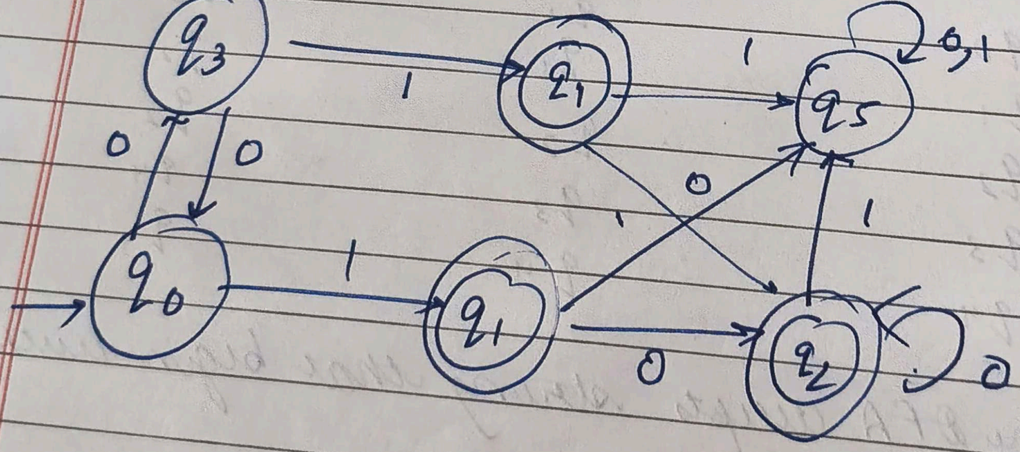
DFA





Ques 22 Minimize the follow DFA

Ans 10 final



final state

$\{q_1, q_2, q_4\}$.

Non-final state

$\{q_0, q_3, q_5\}$

2. Initial state.

$$P_1 = \{\{q_1, q_2, q_4\}, \{q_0, q_3, q_5\}\}$$

final equivalent states

$$q_1 = q_2$$

$$q_0 \equiv q_3$$

4. Minimized state

New DFA states become.

$A = \{q_0, q_3\}$

$B = \{q_1, q_2\}$

$C = \{q_4\}$

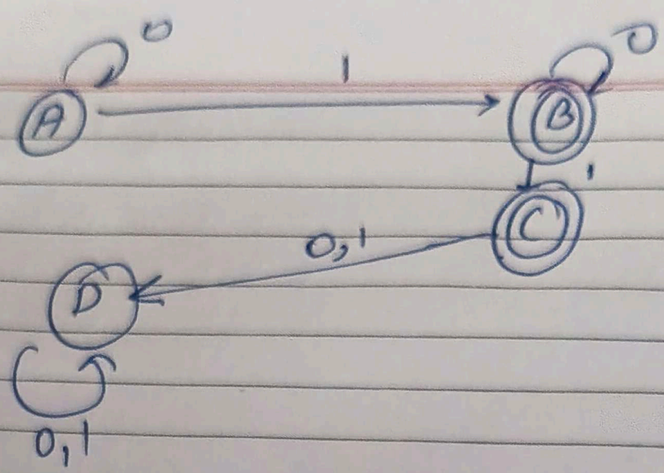
$D = \{q_5\}$

Minimized DFA

state	Input 0	1
A	A	B
B	B	C
C	D	D
D	D	D

start state = A

final = B, C



Ques3) Convert Moore machine to Mealy machine

Ans Moore machine output

state	output
q_0	0
q_1	0
q_2	0
q_3	1
q_4	1

Conversion Rule,

Moore Machine, O/P is associate with states

Mealy machine, O/P is associated with transition

Mealy Machine transition table

present state	Input	Next State	O/P
q_0	0	q_1	0
q_0	1	q_2	0
q_1	0	q_1	0
q_1	1	q_3	1
q_2	0	q_4	1
q_2	1	q_2	0
q_3	0	q_1	0
q_3	1	q_4	1
q_4	0	q_2	0
q_4	1	q_3	1

Ques 12 Explain chomsky classification of lang. taking suitable eg of each classification?

Ans chomsky classification divides formal lang into four types

1. Type 0 :- unrestricted Grammar
   Most general grammer
   Recognized by turing Machine

form $\alpha \longrightarrow \beta$

eg. $L = \{a^n b^n c^n \mid n \geq 1\}$

2. Type -1- Context sensitive Grammer

Rules depend on context
$\alpha A \beta \rightarrow \alpha \gamma \beta$
Recognized by Linear Bounded Automata
$L = \{a^n b^n c^n\}$

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2. Context free.

Rules $A \rightarrow \gamma$

Recognized by pushdown automata

Grammar eg: $L = \{a^n b^n\}$.

$S \longrightarrow aSb / \epsilon$

3. Regular Grammar

simplest Grammar

$A \longrightarrow aB / a$

Recognized by finite automata

eg: $L = \{a^+ b^*\}$

Chomsky.

Regular $\subset$ Context free $\subset$ Context sensitive $\subset$ unrestricted.

Ques 5. Convert NDFA to DFA.

Ans states $\{a, b, c, d, e\}$.

start state a

final state e

NFDA Transition Table.

state	0	1
a	a, c, b, d, e	d, e
b	c	e
c	a	b
d	e	–
e	–	–

subset construction method.

{a}.

DFA Transition table.

DFA	0	1
{a}	{a, b, c, d, e}	{d, e}
{a, b, c, d, e}	{a, b, c, d, e}	{b, d, e}
{d, e}	{e}	ϕ
{e}	ϕ	ϕ
ϕ	ϕ	ϕ

final states are any states containing e

DFA